Quality - LESSON LEARNED CARD

Flexible came out from brush

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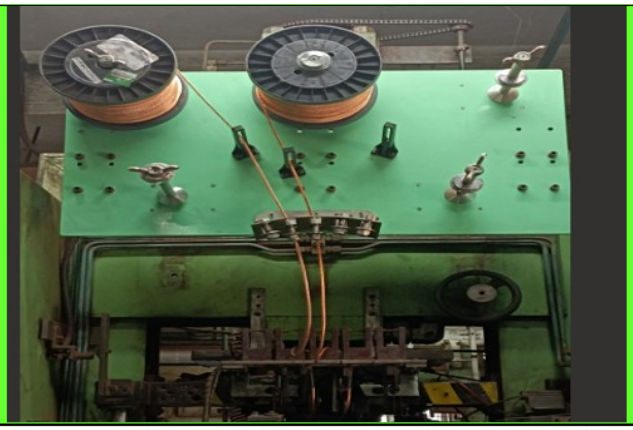
Lesson #	**5494226738**	Severity	**High**
PL.	**Brushes**	App.	**Starter motor**
Customer:	**SEG AUTOMOTIVE**	Prod. Family	**Brush**
Plant	**Chennai**	Part number	**F002G20730 (11368-0)**
Detection	**Line inspection**	Product type	**Brush Starter dual layer**

Problem description:

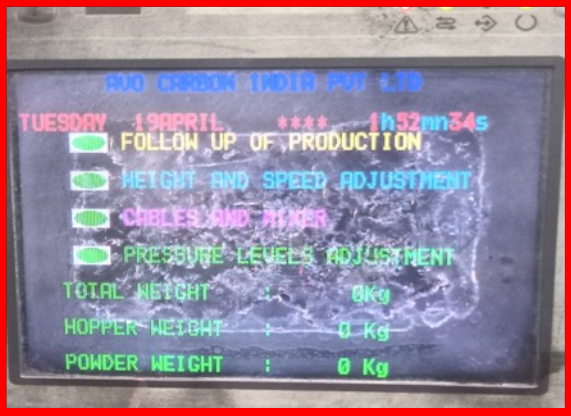
Flexible detached from brush easily and found Pull strength low @ pilling process

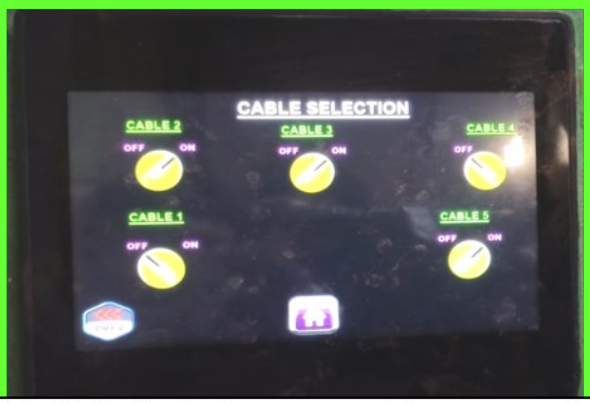
Good Part	Bad Part

Name	Detailled Cause description	Solution description	Conclusion	Process	Origin	File (PDCA/8D analysis)
Situation Befor			Situation After			



separate puller unit placed for each cavity flexible spool and operated by electic motor with sensor unit.





Cable (Flexible) selection incorporate with PLC system if not operate any unit stop machine with alarm.

The puller unit design modified each flexible unit routed with individual puller unit with flex roller sensor for each cavity, spool selection incorporated with each cavity puller unit and operated with an electric motor, this was updated with machine PLC system if any puller unit or the motor not working or pulls back the wire than the machine can stop automatically with warning alarm by sensor.

Distribution to: BRUSH : GERMANY - POITIERS - TIANJIN - CHENNAI	
Plant : CHOKES : TUNISIA - MEXICO - ANHUI - KHUNSHAN - CHENNAI	
Befor	**After**

Plant : BRUSH : GERMANY - POITIERS - TIANJIN - CHENNAI	
Befor	**After**
Plant : SEALS : AMIENS - CHENNAI - TUNISIA - MEXICO	
Befor	**After**
Plant : ASSEMBLIES : TUNISIA - MEXICO - ANHUI - KUNSHAN	
Befor	**After**